SnpEff: Variant analysis

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Summary

Genome	TB_h37rv
Date	2026-10-15 02:48
SnpEff version	`SnpEff 5.4c (build 2026-02-23 07:25), by Pablo Cingolani`
Command line arguments	`SnpEff  TB_h37rv Variant/variant.pass.renamed.vcf`
Warnings	3,818
Errors	0
Number of lines (input file)	1,133
Number of variants (before filter)	1,133
Number of non-variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	1,133
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	0
Number of annotations	11,201
Genome total length	4,411,532
Genome effective length	4,411,532
Variant rate	1 variant every 3,893 bases

Variants rate details

Chromosome	Length	Variants	Variants rate
AL123456.3	4,411,532	1,133	3,893
Total	**4,411,532**	**1,133**	**3,893**

Number variants by type

Type	Total
SNP	1,035
MNP	0
INS	50
DEL	48
MIXED	0
INV	0
DUP	0
CNV	0
BND	0
INTERVAL	0
Total	1,133

Number of effects by impact

Type (alphabetical order)	Count	Percent
HIGH	54	0.482%
LOW	359	3.205%
MODERATE	577	5.151%
MODIFIER	10,211	91.162%

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	552	60.131%
NONSENSE	7	0.763%
SILENT	359	39.107%

Missense / Silent ratio: 1.5376

Number of annotaitons and region counts

Annotation

Type (alphabetical order)	Count	Percent
conservative_inframe_deletion	14	0.125%
conservative_inframe_insertion	8	0.071%
disruptive_inframe_deletion	2	0.018%
disruptive_inframe_insertion	2	0.018%
downstream_gene_variant	4,955	44.221%
frameshift_variant	46	0.411%
intergenic_region	146	1.303%
intragenic_variant	4	0.036%
missense_variant	551	4.917%
splice_region_variant	2	0.018%
start_lost	1	0.009%
stop_gained	8	0.071%
stop_lost	1	0.009%
stop_retained_variant	1	0.009%
synonymous_variant	358	3.195%
upstream_gene_variant	5,106	45.569%

Region

Type (alphabetical order)	Count	Percent
DOWNSTREAM	4,955	44.237%
EXON	989	8.83%
INTERGENIC	146	1.303%
SPLICE_SITE_REGION	1	0.009%
TRANSCRIPT	4	0.036%
UPSTREAM	5,106	45.585%

Quality:

Min	4
Max	228
Mean	208.042
Median	225
Standard deviation	50.743
Values	4,8,13,20,26,27,28,29,30,31,35,37,38,39,40,41,44,46,47,49,50,51,53,56,60,61,62,64,68,71,78,80,82,83,88,90,92,97,99,100,102,103,104,112,120,124
Count	3,3,1,1,1,1,1,1,26,2,1,1,7,3,2,1,2,2,1,1,1,1,2,1,3,2,1,1,2,1,1,2,2,2,1,2,1,1,1,1,1,1,2,1,2,1,2,2,1,1,3,2,1,1,1,1,1,2,2,1,2,1,2,1,1,1,2,1,1,2,1,2,1,3,3,2,5,1,1,

Insertions and deletions length:

Min	0
Max	36
Mean	2.469
Median	1
Standard deviation	5.225
Values	0,1,2,3,4,5,8,9,14,17,20,36
Count	28,50,2,1,5,2,2,2,1,3,1,1

Base changes (SNPs)

	A	C	G	T
A	0	43	157	10
C	56	0	67	153
G	176	84	0	61
T	12	162	54	0

Ts/Tv (transitions / transversions)

Note: Only SNPs are used for this statistic.
Note: This Ts/Tv ratio is a 'raw' ratio (ratio of observed events).

Transitions	648
Transversions	387
Ts/Tv ratio	1.6744

All variants:

```
Sample ,Alignment/alignment.markdup.bam,Total
Transitions ,648,648
Transversions ,387,387
Ts/Tv ,1.674,1.674
```

Only known variants (i.e. the ones having a non-empty ID field):

```
No results available (empty input?)
```

Allele frequency



Min	50
Max	50
Mean	50
Median	50
Standard deviation	0
Values	50
Count	1133

Allele Count



Min	1
Max	1
Mean	1
Median	1
Standard deviation	0
Values	1
Count	1133

Hom/Het per sample



```
Sample_names , Alignment/alignment.markdup.bam
Reference , 0
Het , 1133
Hom , 0
Missing , 0
```

Codon changes

How to read this table:
- Rows are reference codons and columns are changed codons. E.g. Row 'AAA' column 'TAA' indicates how many 'AAA' codons have been replaced by 'TAA' codons.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	-	AAA	AAC	AAG	AAT	ACA	ACC	ACG	ACT	AGA	AGC	AGG	AGT	ATA	ATC	ATG	ATT	CAA	CAC	CAG	CAT	CCA	CCC	C
-			1	1				1			1	1			3			1			1		1	
AAA	1			3						1														
AAC	8	2	**1**	1	5		1				5													
AAG	1	2		**1**	1			2																
AAT			2										1								1			
ACA							1	1																
ACC	7		1			2		6	4						6								1	
ACG	2			3		2	5	**1**				1				3								
ACT							3	1	**1**															
AGA											1													
AGC	3		1				2			1			3											
AGG																								
AGT	1				1						3													
ATA						1									3	3								
ATC	3		1				3				1					2	4							
ATG	5							5						3		**1**								
ATT									1						2	1								
CAA	2	1																**1**	2	5		2		
CAC			1																**3**	2	4			
CAG				7														2			1			
CAT																		1	4	1				
CCA																		3					2	
CCC	1						1												1					

	-	AAA	AAC	AAG	AAT	ACA	ACC	ACG	ACT	AGA	AGC	AGG	AGT	ATA	ATC	ATG	ATT	CAA	CAC	CAG	CAT	CCA	CCC	C
CCG	2																			3		7	2	
CCT									1														3	
CGA										2								1				2		
CGC	5																		6				3	
CGG	1											1								7				
CGT	1												1								2			
CTA																						1		
CTC	2														1								4	
CTG																3								
CTT																					1			
GAA		1																1						
GAC	4		5																1					
GAG	3			3																3				
GAT					1																			
GCA	1					4																		
GCC	14						9																1	
GCG	10							4																
GCT	2								3															
GGA	3																							
GGC	30										7													
GGG	6											2												
GGT	3												2											
GTA																								
GTC	2														7									
GTG	1															7								
GTT																								
TAA																								
TAC	3		1																1					
TAG																								
TAT																								
TCA																								
TCC																							2	
TCG	3																							
TCT																								
TGA																								
TGC																								
TGG	1																							
TGT													1											
TTA	1																							
TTC																								
TTG	2																							
TTT	1																							

Amino acid changes

How to read this table:
- Rows are reference amino acids and columns are changed amino acids. E.g. Row 'A' column 'E' indicates how many 'A' amino acids have been replaced by 'E' amino acids.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	*	-	?	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
*	**1**																						1
-			25	8	1	5	4		13	1	3	1	1		1	3	1	2	2	1	4		
?																							
A		27		**80**		3	7		7							4			6	20	19		
C	1							1	1									5	5			1	1
D		4		1		**13**	5		5	1					6						1		4
E	2	3		2		4	**10**		7			4				1	4				2		
F		1						**6**					6						2		1		
G	1	42		9	3	12	5		**46**									5	9		8		
H						2				**11**			2		1		4	2					3
I		3						1			**9**			6	1				1	5	8		
K		2					9					**6**			1			1		2			
L		5						4		1	1		**40**	3		12		8	2		8	1	
M		5									3		2	**1**						5	6		
N		8				7				1		3			**8**				6	1			
P		3		4						1			10			**34**	6	3	7	2	1		

	*	-	?	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
Q	1	2					1			3		8	1			4	**8**	7					
R	1	7			4				11	8			7			9	8	**29**	2			5	
S		7		4	3				15				4		2	6		3	**17**	2			1
T		9		30		1					6	3		3	1	1		1		**27**			1
V		3		12		1		3	4		7		11	7							**33**		
W		1			1													2					
Y	2	3			4	2				1					1				1				**8**

Variants by chromosome



```
AL123456.3, Position,0,100000,200000,300000,400000,500000,600000,700000,800000,900000,1000000,1100000,1200000,1300000,1400000,150000
AL123456.3,Count,48,34,21,42,39,21,20,15,28,29,31,23,21,39,18,25,24,25,20,24,29,28,28,35,15,29,22,27,24,18,22,20,25,14,25,16,13,29,35,14,27,19,30,39
```

Details by gene

Here you can find a tab-separated table.